



**COMPETENCY STANDARD**

**FOR**

**SEWING MACHINE OPERATION (LINGERIE)**

**(RMG SECTOR)**

**Skills for Industry Competitiveness and Innovation Program (SICIP)**  
**Finance Division, Ministry of Finance**



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The Competency Standards for Sewing Machine Operation (Lingerie) in the RMG sector serves as a foundational resource for designing curricula, creating teaching and learning materials, and developing assessment tools. It is also integral to providing training that aligns with industry demands, ensuring that individuals who meet the established standards through assessment are prepared for relevant job roles.

This document was initially developed under the **Skills for Employment Investment Program (SEIP)** and later revised with input from industry experts specializing in the occupation. It is utilized for training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. Ownership of this document resides with the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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## INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

**The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.**

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

## OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

## Units and elements At a Glance

**Generic competencies (30 hrs.)**

| <b>Code</b>         | <b>Unit of competency</b>                                               | <b>Elements of competency</b>                                                                                                                                                                                                             | <b>Duration (hours)</b> |
|---------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| SICIP-RMG-LSMO-01-G | Apply occupational health and safety (OH&S) practices in the workplace. | <ol style="list-style-type: none"> <li>1. Identify OHS policies and procedures.</li> <li>2. Apply personal health and safety.</li> <li>3. Report hazards and risks.</li> <li>4. Respond to emergencies.</li> </ol>                        | <b>10</b>               |
| SICIP-RMG-LSMO-02-G | Perform computations using basic mathematical concepts.                 | <ol style="list-style-type: none"> <li>1. Identify calculation requirements in the workplace.</li> <li>2. Select appropriate mathematical methods for the calculation.</li> <li>3. Use tool/instrument to perform calculation.</li> </ol> | <b>10</b>               |
| SICIP-RMG-LSMO-03-G | Work in a self-directed team.                                           | <ol style="list-style-type: none"> <li>1. Identify team goals and processes.</li> <li>2. Communicate and cooperate with team members.</li> <li>3. Work as a team member.</li> <li>4. Solve problems as a team member.</li> </ol>          | <b>10</b>               |
| <b>Total hours</b>  |                                                                         |                                                                                                                                                                                                                                           | <b>30</b>               |

**Sector-specific competencies (20 hrs.)**

| <b>Code</b>         | <b>Unit of competency</b> | <b>Elements of competency</b>                                                                                                                                                        | <b>Duration (hours)</b> |
|---------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| SICIP-RMG-LSMO-01-S | Apply green practices.    | <ol style="list-style-type: none"> <li>1. Interpret green concepts.</li> <li>2. Minimize resource use in the workplace.</li> <li>3. Implement waste management practices.</li> </ol> | <b>10</b>               |
| SICIP-RMG-LSMO-02-S | Apply quality procedures. | <ol style="list-style-type: none"> <li>1. Identify quality procedures.</li> <li>2. Follow quality procedures.</li> <li>3. Maintain standard procedures.</li> </ol>                   | <b>10</b>               |
| <b>Total hours</b>  |                           |                                                                                                                                                                                      | <b>20</b>               |

### Occupation-specific competencies (222 hrs.)

| Code                | Unit of competency                                          | Elements of competency                                                                                                                                                                                                                                                                                        | Nominal hours |
|---------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| SICIP-RMG-LSMO-01-O | Demonstrate knowledge on lingerie sewing machine operation. | <ol style="list-style-type: none"> <li>1. Identify types of lingerie.</li> <li>2. Interpret materials for lingerie.</li> <li>3. Identify sewing machine and major parts.</li> <li>4. Recognize sewing machine attachments.</li> </ol>                                                                         | 16            |
| SICIP-RMG-LSMO-02-O | Prepare for lingerie sewing operation.                      | <ol style="list-style-type: none"> <li>1. Follow OHS.</li> <li>2. Arrange tools and materials.</li> <li>3. Adjust sewing machine.</li> <li>4. Set attachments.</li> </ol>                                                                                                                                     | 16            |
| SICIP-RMG-LSMO-03-O | Perform sewing machine operation.                           | <ol style="list-style-type: none"> <li>1. Operate single needle lock stitch machine.</li> <li>2. Operate double needle lockstitch machine.</li> <li>3. Operate over lock machine.</li> <li>4. Operate flat lock machine.</li> <li>5. Operate zigzag machine.</li> <li>6. Operate bar tack machine.</li> </ol> | 108           |
| SICIP-RMG-LSMO-04-O | Perform lingerie product sewing.                            | <ol style="list-style-type: none"> <li>1. Collect bra and brief components.</li> <li>2. Carry out brief stitching.</li> <li>3. Carry out bra stitching.</li> </ol>                                                                                                                                            | 62            |
| SICIP-RMG-LSMO-05-O | Maintain optimum production.                                | <ol style="list-style-type: none"> <li>1. Maintain production target and efficiency.</li> <li>2. Maintain products quality.</li> <li>3. Perform basic maintenance.</li> </ol>                                                                                                                                 | 20            |
| <b>Total Hour</b>   |                                                             |                                                                                                                                                                                                                                                                                                               | <b>222</b>    |



## COMPETENCY STANDARD: SEWING MACHINE OPERATION (LINGERIE)

### Generic Competencies

|                                                                                                                                                                                                                                                                                                                                                   |                                     |                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE</b>                                                                                                                                                                                                                                         | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-01-G |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety, reporting hazards and risks and responding to emergencies. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                  | Performance Criteria                                                                                                                                                                                                                       |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify OHS policies and procedures | 1.1 <b><u>OHS policies</u></b> and safe operating procedures are interpreted<br>1.2 Safety signs and symbols are identified and followed.<br>1.3 Response, evacuation procedures and other contingency measures are interpreted correctly. |
| 2. Apply personal health and safety     | 2.1 OHS policies and procedures are interpreted in the workplace including <b><u>personal protective equipment (PPE)</u></b> .<br>2.2 Common health issues are recognized.<br>2.3 Common safety issues are identified.                     |
| 3. Report hazards and risks             | 3.1 Hazards and risks are identified.<br>3.2 Hazards and risks assessment and controls are interpreted                                                                                                                                     |
| 4. Respond to emergencies               | 4.1 Respond to alarms and warning devices.<br>4.2 <b><u>Emergency response plans and procedures</u></b> are responded to.<br>4.3 <b><u>First aid procedures</u></b> during emergency situations are identified.                            |

### Range of variables:

| Variables                        | Range (may include but not limited to)                                                                         |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1. OHS policies                  | 1.1 Organizational OHS policies<br>1.2 International OHS requirements<br>1.3 Fire safety rules and regulations |
| 2. Personal protective equipment | 2.1 Safety glasses<br>2.2 Ear plugs<br>2.3 Gloves<br>2.4 Apron                                                 |

|                                            |                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | 2.5 Helmet<br>2.6 Mask<br>2.7 Safety shoes                                                                                                    |
| 3. Emergency response plans and procedures | 3.1 Firefighting procedures<br>3.2 Earthquake response procedures<br>3.3 Emergency response plans and procedures<br>3.4 Medical and first aid |
| 4. First aid procedure                     | 4.1 Washing of open wound<br>4.2 Washing chemically infected area<br>4.3 Applying bandage<br>4.4 Taking appropriate medicine                  |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1 Workplace OHS policies and procedures<br>1.2 Work safety procedures<br>1.3 Emergency response procedures:<br>1.3.1 Firefighting<br>1.3.2 Earthquake response<br>1.3.3 Accident response<br>1.4 Types of hazards (biological, chemical and physical) and their effects<br>1.5 OHS awareness<br>1.6 Personal protective equipment (PPE)<br>1.7 Malfunctions and resolutions |
| 2. Underpinning skill     | 2.1 Identifying OHS policies and procedures<br>2.2 Applying personal health and safety practices<br>2.3 Reporting hazards and risks<br>2.4 Responding to emergencies                                                                                                                                                                                                          |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace           |
| 4. Resource implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Personal protective equipment (PPE)<br>4.3 Firefighting equipment<br>4.4 Emergency response manual<br>4.5 First aid kits<br>4.6 Manuals                                                                                                                                               |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | <p>Assessment required evidence that the candidate:</p> <ul style="list-style-type: none"> <li>1.1 identified OHS policies and procedures</li> <li>1.2 applied personal health and safety practices</li> <li>1.3 reported hazards and risks</li> <li>1.4 responded to emergencies.</li> </ul>                         |
| 2. Methods of assessment          | <p>Competency should be assessed by:</p> <ul style="list-style-type: none"> <li>2.1 Written test</li> <li>2.2 Practical Demonstration</li> <li>2.3 Oral Questioning</li> <li>2.4 Portfolio (Optional)</li> </ul>                                                                                                      |
| 3. Context of assessment          | <ul style="list-style-type: none"> <li>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</li> <li>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li> </ul> |

|                                                                                                                                                                                                                                                                                                                                                                      |                                     |                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>PERFORM COMPUTATIONS USING BASIC MATHEMATICAL CONCEPTS</b>                                                                                                                                                                                                                                                                          | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-02-G |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to perform computations using basic mathematical concepts. It specifically includes the tasks of identifying calculation requirements in the workplace, selecting appropriate mathematical methods for the calculation and using tools/ instruments to perform calculation. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                                          | Performance Criteria                                                                                                                                                                                                                         |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify calculation requirements in the workplace           | 1.1 Job requirements are identified.<br>1.2 <b><u>Measurements</u></b> are selected in accordance with job requirement<br>1.3 Calculation requirements are identified from <b><u>workplace information.</u></b>                              |
| 2. Select appropriate mathematical methods for the calculation. | 2.1. Mathematical methods are identified<br>2.2. <b><u>Appropriate method</u></b> is selected to carry out the calculation requirements<br>2.3. Tolerance and clearance limits are identified and adjusted according to the job requirements |
| 3. Use tool/instrument to perform calculations                  | 3.1 Work instructions are confirmed and applied to the job in hand<br>3.2 Materials to be measured are identified as per job specification<br>3.3 Appropriate tool/ instrument is selected based on materials to be measured                 |

### Range of Variables

| Variable                 | Range (May Include but not limited to):                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1. Measurements          | 1.1 Length<br>1.2 Width<br>1.3 Weight<br>1.4 Tolerance                                                                          |
| 2. Workplace information | 2.1. Job order<br>2.2. Trims card<br>2.3. Working drawing<br>2.4. Verbal instructions<br>2.5. Written Instruction               |
| 3. Appropriate method    | 3.1 Addition<br>3.2 Subtraction<br>3.3 Division<br>3.4 Multiplication<br>3.5 Conversion<br>3.6 Percentage and ratio calculation |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| 4. Tool/ Instrument | 4.1 Calculator<br>4.2 Scale<br>4.3 Measuring tape<br>4.4 Marker |
|---------------------|-----------------------------------------------------------------|

#### **Curricular Content Guide:**

|                           |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Numerical concept<br>1.2 Basic mathematical methods such as addition, subtraction, multiplication and division and percentage.<br>1.3 Mathematical language, symbols and terminology.<br>1.4 Measuring units                                                                                                                                                    |
| 2. Underpinning Skills    | 2.1. Selecting measurements<br>2.2. Selecting appropriate methods<br>2.3. Using tool/instrument<br>2.4. Adding numbers<br>2.5. Subtracting numbers<br>2.6. Multiplying numbers.<br>2.7. Dividing numbers.                                                                                                                                                           |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Required Tool/ Instrument<br>4.3 Pens<br>4.4 Papers<br>4.5 Work books<br>4.6 Learning manuals                                                                                                                                                                               |

#### **Assessment Evidence Guide:**

|                                   |                                                                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 identified calculation requirements from workplace information.<br>1.2 selected appropriate method to carry out the calculation requirements.<br>1.3 completed calculations using appropriate tools/ instruments. |
| 2. Methods of Assessment          | Methods of assessment may include but not limited to:                                                                                                                                                                                                                     |

|                          |                                                                                                                                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                                                               |
| 3. Context of Assessment | 3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert |

|                                                                                                                                                                                                                                                                                                                        |                                     |                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>WORK IN A SELF-DIRECTED TEAM</b>                                                                                                                                                                                                                                                      | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-03-G |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to work in a self-directed team. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                          | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify team goals and processes            | 1.1 Team goals and collaborative decision-making processes are identified.<br>1.2 Roles and responsibilities of team members are identified<br>1.3 Relationships within team and with other workers are identified                                                                                                                                                                                                                                              |
| 2. Communicate and cooperate with team members. | 2.1. Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives<br>2.2. Formal and informal <b><u>forms of communication</u></b> are used effectively to support team achievement.<br>2.3. Diversity is respected and valued in team functioning.<br>2.4. Views and opinions of other team members are understood and valued.<br>2.5. Workplace terminology is used correctly to assist communication |
| 3. Work as a team member                        | 3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team<br>3.2 Tasks are performed in accordance with organizational and team requirements, specifications and workplace procedures.<br>3.3 Team member's support with other members are made to ensure team achieves goals, awareness and requirements.<br>3.4 Agreed reporting lines are followed using standard operating procedure.              |
| 4. Solve problems as a team member              | 4.1 Current and potential problems faced by team are identified<br>4.2 A solution to the problem is identified<br>4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated                                                                                                                                                                                                                                                  |

## Range of Variables

| Variable                  | Range (May Include but not limited to):                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Forms of communication | 1.1 Agenda<br>1.2 Simple reports such as progress and incident reports.<br>1.3 Job sheets/Job card.<br>1.4 Operational manuals.<br>1.5 Brochures and promotional material.<br>1.6 Visual and graphic materials.<br>1.7 OSH information.<br>1.8 Signs. |

## Curricular Content Guide:

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Team goals and collaborative decision-making processes<br>1.2 Roles and responsibilities of team members<br>1.3 Relationships within team and with other workers<br>1.4 Effective interpersonal skills to interact with team members<br>1.5 Effective formal and informal forms of communication<br>1.6 Value of diversity in team functioning.<br>1.7 Correct use of workplace terminology<br>1.8 Team's duties, responsibilities, authorities, objectives and task requirements<br>1.9 Support mechanism to other members of team to ensure achievements of goals.<br>1.10 Methods of identifying current and potential problems faced by a team<br>1.11 Effectively problems solving methods and evaluation of outcomes |
| 2. Underpinning Skills    | 2.1. Identifying team goals and collaborative decision-making processes<br>2.2. Identifying roles and responsibilities of team members<br>2.3. Identifying relationships within team and with other workers<br>2.4. Using effective interpersonal skills to interact with team members and to contribute to activities and objectives<br>2.5. Using formal and informal forms of communication<br>2.6. Understanding and valuing views and opinions of other team members<br>2.7. Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures.<br>2.8. Supporting other members of the team to ensure team achieves goals, awareness and requirements.                   |



|                           |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Pens<br>4.3 Papers<br>4.4 Work books<br>4.5 Learning manuals                                                                                                                                                                                                                |

#### **Assessment Evidence Guide:**

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br><br>1.1 Identified Roles and responsibilities of team members<br>1.2 Identified Relationships within team and with other workers<br>1.3 Correctly used Workplace terminology to assist communication<br>1.4 Identified and clarified the duties, responsibilities, authorities, objectives and task requirements with team<br>1.5 Identified Current and potential problems faced & solution to the problem<br>1.6 Evaluated outcome of the implemented solution |
| 2. Methods of Assessment          | Methods of assessment may include but not limited to:<br><br>2.1. Written test<br>2.2. Practical Demonstration<br>2.3. Oral questioning<br>2.4. Portfolio (Optional)                                                                                                                                                                                                                                                                                                                                                 |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.                                                                                                                                                                                                                                                                  |

## The Sector Specific Competencies

|                                                                                                                                                                                                                                                                                                 |                                     |                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY GREEN PRACTICES IN RMG SECTOR</b>                                                                                                                                                                                                                        | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-01-S |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required for apply green practices in RMG sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                     | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret green concepts                | 1.1 <b><u>Principles of green practices in RMG (Ready-Made Garment)</u></b> manufacturing are explained.<br>1.2 Key terms and symbols in RMG production processes are identified.<br>1.3 <b><u>Sources of environmental impacts</u></b> during RMG manufacturing activities are identified.<br>1.4 RMG manufacturing activities use are explained.<br>1.5 <b><u>Ways of mitigating environmental impacts</u></b> in RMG manufacturing are explained. |
| 2. Minimize resources use in the workplace | 2.1 Water, energy, and raw material consumption in RMG manufacturing are documented.<br>2.2 <b><u>Recyclable and non-recyclable items</u></b> in RMG production processes are identified.<br>2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented.<br>2.4 Sustainable alternatives in RMG manufacturing are explored and applied.                                                                                      |
| 3. Implement waste management practices    | 3.1 <b><u>Different types of waste in RMG manufacturing</u></b> are identified.<br>3.2 Hazardous waste in RMG production is disposed of according to environmental regulations.<br>3.3 Green habits to reduce waste in personal and professional life are practiced.                                                                                                                                                                                 |

### Range of Variables

| Variable                                                     | Range (May include but not limited to):                                                                                                                                                                                           |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Principles of green practices in RMG (Ready-Made Garment) | 1.1 Reducing energy consumption in RMG manufacturing processes.<br>1.2 6R for waste management in RMG production:<br>1.2.1 Refuse unnecessary materials and processes.<br>1.2.2 Reduce resource consumption and waste generation. |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>1.2.3 Reuse materials within the production process.</p> <p>1.2.4 Recycle materials for future use in RMG manufacturing.</p> <p>1.2.5 Recover usable resources from waste.</p> <p>1.2.6 Repair damaged products or materials instead of discarding them.</p> <p>1.3 Use of sustainable materials in RMG production with low environmental impact.</p> <p>1.4 Recycling of materials used in the RMG manufacturing process.</p> <p>1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.</p>                                                                                           |
| 2. Sources of environmental impacts              | <p>2.1 Air Pollution in RMG Manufacturing</p> <p>2.2 Water Pollution in RMG Manufacturing</p> <p>2.3 Soil Degradation in RMG Manufacturing</p> <p>2.4 Noise Pollution in RMG Manufacturing</p> <p>2.5 Waste Generation in RMG Manufacturing</p> <p>2.6 Resource Depletion in RMG Manufacturing</p>                                                                                                                                                                                                                                                                                                                              |
| 3. Ways of mitigating environmental impacts      | <p>3.1 Utilizing energy-efficient equipment in RMG production processes.</p> <p>3.2 Adopting renewable energy sources for RMG manufacturing operations.</p> <p>3.3 Implementing site protection measures to prevent pollution and reduce environmental.</p> <p>3.4 Using reusable materials in production processes.</p> <p>3.5 Choosing sustainable materials with low environmental impact for garment production.</p> <p>3.6 Using noise-reducing equipment and scheduling production work.</p> <p>3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.</p> |
| 4. Recyclable and non-recyclable items           | <p>4.1 <b>Recyclable Items in RMG Manufacturing</b></p> <p>4.1.1 Metal components.</p> <p>4.1.2 Wooden materials.</p> <p>4.1.3 Fabric scraps and textile fibers.</p> <p>4.1.4 Plastic items, including buttons, tags, and packaging materials.</p> <p>4.2 <b>Non-Recyclable Items in RMG Manufacturing</b></p> <p>4.2.1 Paints, coatings, and dyes.</p> <p>4.2.2 Contaminated materials.</p> <p>4.2.3 Treated wood and other materials.</p>                                                                                                                                                                                     |
| 5. Different types of waste in RMG manufacturing | <p>5.1 Fabric waste from cutting and trimming processes.</p> <p>5.2 Wood waste from packaging or display materials.</p> <p>5.3 Metal waste from trims, buttons, zippers, and other hardware.</p> <p>5.4 Textile waste, including scraps from garment</p>                                                                                                                                                                                                                                                                                                                                                                        |

|  |                                                                                                                                                                                                                                                                         |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>production and defective products.</p> <p>5.5 Plastic waste, such as packaging materials, labels, and plastic buttons.</p> <p>5.6 Solid waste from chemical treatments or dyeing processes.</p> <p>5.7 Packaging waste from raw materials and finished products.</p> |
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### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <p>1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing.</p> <p>1.2 Sources of environmental impacts.</p> <p>1.3 Ways of mitigating environmental impacts</p> <p>1.4 Recyclable and non-recyclable items</p> <p>1.5 Different types of waste in RMG manufacturing</p> <p>1.6 Hazardous waste in RMG production</p> <p>1.7 Green habits to reduce waste</p>                                               |
| 2. Underpinning Skills    | <p>2.1 Explaining principles of green practices in RMG manufacturing</p> <p>2.2 Identifying sources of environmental impacts</p> <p>2.3 Explaining re-cyclable and non-recyclable items</p> <p>2.4 Minimizing resources use in the workplace</p> <p>2.5 Identifying different types of waste in RMG manufacturing</p> <p>2.6 Disposing hazardous waste in RMG production</p> <p>2.7 Practicing green habits to reduce waste</p> |
| 3. Underpinning Attitudes | <p>3.1 Commitment to occupational health and safety</p> <p>3.2 Promptness in carrying out activities</p> <p>3.3 Sincere and honest to duties</p> <p>3.4 Environmental concerns</p> <p>3.5 Eagerness to learn</p> <p>3.6 Tidiness and timeliness</p> <p>3.7 Respect for rights of peers and seniors in the workplace</p> <p>3.8 Communication with peers, subordinates and seniors in the workplace</p>                          |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <p>4.1 Workplace (simulated or actual)</p> <p>4.2 Projector</p> <p>4.3 Stationery</p> <p>4.4 Learning manual</p>                                                                                                                                                                                                                                                               |

### Assessment Evidence Guide

|                                   |                                                  |
|-----------------------------------|--------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate: |
|-----------------------------------|--------------------------------------------------|

|                          |                                                                                                                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 1.1 interpreted green concepts.<br>1.2 minimizing resources use in the workplace.<br>1.3 implementing waste management practices.                                                                                                                             |
| 2. Methods of Assessment | Competency should be assessed by:<br><br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                  |
| 3. Context of Assessment | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

|                                                                                                                                                                                                                                                                   |                                     |                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY QUALITY PROCEDURES</b>                                                                                                                                                                                                     | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-02-S |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to apply quality procedures. It specifically includes the tasks of identifying quality procedures, following quality procedures and maintaining standard procedures. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency         | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify quality procedures | 1.1. <b><u>Manuals</u></b> and guidelines for RMG production processes are collected as per sample specifications.<br>1.2. The importance of manuals in ensuring quality standards is recognized.<br>1.3. Instructions and procedures for quality control are identified.<br>1.4. Required information is collected from manuals.<br>1.5. Performance measurement systems are identified.                                                                            |
| 2. Follow quality procedure    | 2.1 Instructions and procedures are followed strictly and duties are performed in accordance with demand of <b><u>quality improvement system</u></b> .<br>2.2 Concept of supplying product or service to meet the <b><u>customer quality requirements</u></b> is understood and accordingly applied.<br>2.3 Conformance to specifications is ensured.<br>2.4 Defects are detected and reported to the relevant authority according to standard operating procedures. |
| 3. Maintain standard procedure | 3.1 Performance is assessed at regular interval.<br>3.2 Specifications and standard operating procedures are established.<br>3.3 Quality of product is checked and verified.<br>3.4 Quality control and quality assurance system procedures for each job are followed.<br>3.5 Conformance to specification is ensured in every case at all situations.                                                                                                               |

### Range of Variables

| Variable   | Range (May include but not limited to):                                                                                                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Manuals | 1.1 Buyers' specification manual<br>1.2 Compliance manual<br>1.3 Maintenance procedure manual<br>1.4 Periodic maintenance manual<br>1.5 Quality manual |

|                                  |                                                                                                                             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|                                  | 1.6 Signs and symbols, instruction manuals                                                                                  |
| 2. Quality improvement system.   | 2.1. Quality inspection<br>2.2. Testing<br>2.3. Quality control.<br>2.4. Quality assurance<br>2.5. Total Quality Management |
| 3. Customer quality requirements | 3.1. Performance<br>3.2. Features<br>3.3. Reliability<br>3.4. Conformance<br>3.5. Aesthetics<br>3.6. Durability             |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Importance of maintaining quality<br>1.2 quality, quality assurance, quality control, quality inspection, quality improvement and total quality control.<br>1.3 Process and procedures for improving and maintaining quality<br>1.4 Procedures for addressing defects.<br>1.5 Record keeping within the quality improvement system in workplace<br>1.6 Factors, which affect successful implementation of the quality systems and procedures.      |
| 2. Underpinning Skills    | 2.1. Maintaining good quality.<br>2.2. Eliminating poor quality.<br>2.3. Understanding the meaning of the key terms - quality, quality assurance, quality control, quality inspection, quality improvement and total quality control.<br>2.4. Improving and maintaining quality.<br>2.5. Addressing defects and procedures.<br>2.6. Recording within the quality improvement system in workplace.<br>2.7. Implementing quality systems and procedures. |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in workplace<br>3.8 Communication with peers, sub-ordinates and seniors in workplace                                                                                           |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)                                                                                                                                                                                                                                                                                                                                                                       |

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|  | 4.2 Map/globe<br>4.3 Projector<br>4.4 Stationery<br>4.5 Learning manual. |
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### Assessment Evidence Guide

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|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 identified quality procedures<br>1.2 followed quality procedure<br>1.3 maintained standard procedure                                                                                                  |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                      |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |



## The Occupation Specific Competencies

|                                                                                                                                                                                                                                                                                                                                                              |                                     |                                          |
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| <b>Unit of Competency:</b><br><b>DEMONSTRATE KNOWLEDGE ON LINGERIE SEWING MACHINE OPERATION</b>                                                                                                                                                                                                                                                              | <b>Nominal Duration:</b><br>16 hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-01-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to demonstrate knowledge on lingerie sewing machine operation. It specifically includes the tasks of identifying types of lingerie, interpreting materials for lingerie, identifying sewing machine and major parts and recognizing sewing machine attachments. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                      | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify types of lingerie.              | 1.1 The different <b><u>types of lingerie</u></b> are described.<br>1.2 Key features of each lingerie type are understood.<br>1.3 The purpose of each type of lingerie is identified.                                                                                                                                                                                                                             |
| 2. Interpret materials for lingerie.        | 2.1. The different <b><u>materials for lingerie</u></b> are explained.<br>2.2. The characteristics of each material are recognized.<br>2.3. The suitability of each material for various lingerie types is understood.                                                                                                                                                                                            |
| 3. Identify sewing machine and major parts. | 3.1 Different <b><u>types of sewing machines</u></b> are described.<br>3.2 <b><u>Major parts</u></b> of a sewing machine are recognized.<br>3.3 Function of each major part is understood.<br>3.4 <b><u>Safety guards</u></b> are identified as per job requirement.                                                                                                                                              |
| 4. Recognize sewing machine attachments.    | 4.1 Various <b><u>sewing machine attachments</u></b> are identified.<br>4.2 The function of each attachment is understood.<br>4.3 The correct attachment for different sewing tasks is identified based on the project requirements.<br>4.4 The process of attaching and detaching each sewing machine attachment is recognized.<br>4.5 Basic maintenance and care for sewing machine attachments are understood. |

### Range of Variables

| Variable             | Range<br>May Include but not limited to:                                                                                 |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1. Types of lingerie | 1.1 Bra<br>1.1.1 Wire mould Bra<br>1.1.2 Padded bra<br>1.1.3 Wireless bra<br>1.2 Bikini<br>1.3 Panty<br>1.4 Ladies brief |

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|                           | 1.5 Swimwear<br>1.6 Bodysuit<br>1.7 Bandeau<br>1.8 Sports bra<br>1.9 Thong<br>1.10 Garter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2. Materials for lingerie | 2.1. Fabrics<br>2.1.1 Micro fabric<br>2.1.2 Mesh fabric<br>2.1.3 Single jersey cotton/lycra fabric<br>2.1.4 Poplin fabric<br>2.1.5 Siphon cotton fabric<br>2.1.6 Satin fabric<br>2.1.7 Silk fabric<br>2.1.8 Polyamide<br>2.1.9 Tricot fabric<br>2.1.10 Soft net<br>2.1.11 Fixed net<br>2.2. Sewing thread<br>2.1.1 PP cotton<br>2.1.2 Filament<br>2.1.3 Elastin<br>2.3. Decorative metal<br>2.4. Label<br>2.5. Elastic<br>2.6. Lace<br>2.7. Button<br>2.8. Hook & eye<br>2.9. Ring slider<br>2.10. Buckles<br>2.11. Snape tape<br>2.12. Ribbon<br>2.13. Bow<br>2.14. Drawstring<br>2.15. Wire<br>2.16. Plastic bone<br>2.17. Wire and bone case<br>2.18. Suspender clips |
| 3. Sewing machine         | 3.1 Single needle lock stitch machine<br>3.2 Single needle edge cutter lockstitch machine<br>3.3 Double needle lock stitch machine<br>3.4 Over lock machine<br>3.5 Flat lock machine<br>3.6 Zigzag machine<br>3.7 Bar tack machine<br>3.8 Flat seamer machine<br>3.9 Ultrasonic seamless machine                                                                                                                                                                                                                                                                                                                                                                         |
| 4. Major parts            | 4.1 Feed dog                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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|                               | 4.2 Needles<br>4.3 Needle plate<br>4.4 Needle bar<br>4.5 Thread bar<br>4.6 Tension post assembly<br>4.7 Bobbin case and spool<br>4.8 Thread guide<br>4.9 Pressure adjusting screw<br>4.10 Pressure foot lever<br>4.11 Pressure foot<br>4.12 Hand wheel<br>4.13 Knee lifter<br>4.14 Stitch regulator<br>4.15 Thread wiper/nipper<br>4.16 Needle gauge |
| 5. Safety guards              | 5.1 Motor guard<br>5.2 Eye guard<br>5.3 Belt guard<br>5.4 Needle guard<br>5.5 Paddle mat                                                                                                                                                                                                                                                             |
| 6. Sewing machine attachments | 6.1 Feed<br>6.2 Guide<br>6.3 Folder<br>6.4 Measuring device<br>6.5 Metering device<br>6.6 Chopper<br>6.7 Auto cutting and reinsert                                                                                                                                                                                                                   |

### Curricular Content Guide:

|                           |                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Types and purpose of lingerie.<br>1.2 Materials for lingerie sewing.<br>1.3 Types of sewing machine.<br>1.4 Major parts and function of sewing machine.<br>1.5 Safety guards.<br>1.6 Sewing machine attachments and function.<br>1.7 Process of attaching and detaching.<br>1.8 Maintenance of sewing machine attachment. |
| 2. Underpinning Skills    | 2.1 Categorizing type of lingerie.<br>2.2 Listing characteristics of lingerie materials.<br>2.3 Identifying major parts of a sewing machine.<br>2.4 Categorizing sewing machine attachments.                                                                                                                                  |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn                                                                                                                                     |

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|                          | 3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace                                                                                                                                                          |
| 4. Resource Implications | The following resources must be provided:<br><br>4.1 Workplace (simulated or actual)<br>4.2 Required machine, tools and equipment and facilities appropriate to the process or activities.<br>4.3 Sewing machine attachment and materials are relevant to the proposed activity.<br>4.4 Papers<br>4.5 Work books<br>4.6 Manuals |

#### **Assessment Evidence Guide:**

|                                   |                                                                                                                                                                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br><br>1.1 identified type of lingerie.<br>1.2 identified materials for lingerie.<br>1.3 identified sewing machine and major parts.<br>1.4 recognized sewing machine attachments.                  |
| 2. Methods of Assessment          | Methods of assessment may include but not limited to:<br><br>2.1. Written test<br>2.2. Practical Demonstration<br>2.3. Oral questioning<br>2.4. Portfolio (Optional)                                                                                |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert. |

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| <b>Unit of Competency:</b><br><b>PREPARE FOR LINGERIE SEWING OPERATION</b>                                                                                                                                                                                               | <b>Nominal Duration:</b><br>16 Hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-02-O |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to prepare for lingerie sewing operation. It specifically includes the tasks of following OHS, arranging tools and materials, adjusting sewing machine and setting attachments. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency          | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Follow OSH.                  | 1.1 <b><u>Personal protective equipment (PPE)</u></b> is collected and used as per job requirement.<br>1.2 Occupational safety and health (OSH) are followed as per job requirement.<br>1.3 Machine is cleaned and checked as required.                                                                                                                                                                                     |
| 2. Arrange tools and materials. | 2.1 The necessary <b><u>tools</u></b> and materials for the task are identified.<br>2.2 The correct tools and materials are selected based on the type of fabric.<br>2.3 <b><u>Cut panels</u></b> are received and maintained as per number and bundle.<br>2.4 The materials are received as needed for the task.<br>2.5 Safety measures are considered while handling tools and materials to prevent accidents and damage. |
| 3. Adjust sewing machine.       | 3.1 Safety guards are checked as required.<br>3.2 Needles are checked and adjusted.<br>3.3 Broken needles are replaced as per standard operating procedure (SOP).<br>3.4 Machine parts are checked and adjusted as required.<br>3.5 Tension of the stitches are checked and adjusted as required.<br>3.6 Machine is checked to ensure Stich Per Inch (SPI).                                                                 |
| 4. Set attachments.             | 4.1 Sewing machine attachments are selected and collected as per job requirement.<br>4.2 Sewing machine attachments are set as per job requirement.<br>4.3 Sewing attachments are checked and adjusted.                                                                                                                                                                                                                     |

### Range of Variables

| Variable                               | Range (Includes but not limited to):             |
|----------------------------------------|--------------------------------------------------|
| 1. Personal protective equipment (PPE) | 1.1 Mask<br>1.2 Apron/Uniform<br>1.3 Hand gloves |

|               |                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
|               | 1.4 Scarf/cap<br>1.5 Safety glass                                                                                                      |
| 2. Tools      | 2.1 Scissor<br>2.2 Trimmer<br>2.3 Stitch opener<br>2.4 Tweezer<br>2.5 Screw driver<br>2.6 Needle driver<br>2.7 Allen key<br>2.8 Mallet |
| 3. Cut panels | 3.1 Bra<br>3.1.1 Yoke/Gore<br>3.1.2 Shell cup<br>3.1.3 Wing part<br>3.2 Brief<br>3.2.1 Front part<br>3.2.2 Back part<br>3.2.3 Gusset   |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1. Occupational safety and health (OSH).<br>1.2. Tools and materials for lingerie operation.<br>1.3. Different types of fabric.<br>1.4. Describe cut panels.<br>1.5. Safety guards<br>1.6. Tension of the stitches<br>1.7. Sewing attachments                                                                                                                             |
| 2. Underpinning Skills    | 2.1. Checking and cleaning technique of machine.<br>2.2. Selecting tools and materials.<br>2.3. Checking safety guards<br>2.4. Checking machine parts<br>2.5. Checking tension of the stitch<br>2.6. Selecting sewing attachments                                                                                                                                           |
| 3. Underpinning Attitudes | 3.1. Commitment to occupational health and safety<br>3.2. Promptness in carrying out activities<br>3.3. Sincere and honest to duties<br>3.4. Environmental concerns<br>3.5. Eagerness to learn<br>3.6. Tidiness and timeliness<br>3.7. Respect for rights of peers and seniors in the workplace<br>3.8. Communication with peers, subordinates and seniors in the workplace |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Tools, equipment and facilities appropriate to the process or activities.                                                                                                                                                                                                           |

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|  | 4.3 Materials are relevant to the proposed activity.<br>4.4 Manuals<br>4.5 Work book. |
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### Assessment Evidence Guide

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|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <ul style="list-style-type: none"> <li>1.1 Followed occupational safety and health (OSH).</li> <li>1.2 Identified tools and materials.</li> <li>1.3 Checked safety guards.</li> <li>1.4 Adjusted sewing machine</li> <li>1.5 Checked stitch per inch.</li> <li>1.6 Set sewing machine attachments.</li> </ul> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <ul style="list-style-type: none"> <li>2.1 Written test</li> <li>2.2 Practical Demonstration</li> <li>2.3 Oral Questioning</li> <li>2.4 Portfolio (Optional)</li> </ul>                                                                                                                                                      |
| 3. Context of Assessment          | <ul style="list-style-type: none"> <li>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</li> <li>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li> </ul>                                                 |

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| <b>Unit of Competency:</b><br><b>PERFORM SEWING MACHINE OPERATION</b>                                                                                                                                                                                                                                                                                                          | <b>Nominal Duration:</b><br>108Hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-03-O |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to perform sewing machine operation. It specifically includes the tasks of operating single needle lockstitch machine, operating double needle lockstitch machine, operating over lock machine, operating flat lock machine, operating zigzag machine and operating bar tack machine. |                                     |                                          |

**Elements and Performance Criteria:**

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                      | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Operate single needle lockstitch machine | 1.1 Single needle lockstitch machine is set up and the machine is properly threaded.<br>1.2 Stitch length and tension are adjusted according to the fabric.<br>1.3 Sample fabric/cut piece is positioned under the presser foot, and the machine is started, ensuring smooth and even stitching.<br>1.4 The sewing speed is controlled to maintain consistency and accuracy throughout the stitching process.<br>1.5 During sewing, <b><u>unusual conditions</u></b> (if any) are detected and reported to authority.<br>1.6 Machine is cleaned and maintained according to the manufacturer's guidelines.                                                                                                                                                                                        |
| 2. Operate double needle lockstitch machine | 2.1. Double needle lockstitch machine is set up by ensuring the correct needles and threads.<br>2.2. Machine is threaded properly for double stitching.<br>2.3. Stitch width, length, and tension are adjusted according to the fabric type and stitching requirements.<br>2.4. Sample fabric/cut piece is positioned under both needles and the presser foot.<br>2.5. Machine is operated at a steady speed, ensuring even and consistent stitching on both sides.<br>2.6. Sewing path is monitored to ensure both needles are stitching parallel lines with accurate spacing.<br>2.7. During sewing, unusual conditions (if any) are detected and reported to authority.<br>2.8. The machine is cleaned and maintained according to the manufacturer's instructions to ensure smooth operation. |
| 3. Operate over lock machine                | 3.1 Overlock machine is set up by ensuring the correct thread spools and needles.<br>3.2 Machine is threaded properly for a smooth stitch.<br>3.3 Sample fabric/cut piece is positioned under the presser foot.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>3.4 The machine is operated at a steady speed to create a clean, finished edge on the fabric.</p> <p>3.5 Stitch length and tension are adjusted according to the fabric type to prevent puckering or thread breakage.</p> <p>3.6 During sewing, unusual conditions (if any) are detected and reported to authority.</p> <p>3.7 The machine is cleaned and maintained regularly according to the manufacturer's guidelines for optimal performance.</p>                                                                                                                                                                                          |
| 4. Operate flat lock machine | <p>4.1 Flat lock machine is set up by ensuring the correct needles, threads, and tensions.</p> <p>4.2 Machine is threaded properly to ensure a balanced stitch.</p> <p>4.3 Sample fabric/cut piece is positioned under the presser foot.</p> <p>4.4 Flat lock machine is operated at a consistent speed.</p> <p>4.5 The stitch length and tension are adjusted based on the fabric type and stitch.</p> <p>4.6 During sewing, unusual conditions (if any) are detected and reported to authority.</p> <p>4.7 The machine is cleaned and maintained regularly according to the manufacturer's instructions.</p>                                     |
| 5. Operate zigzag machine    | <p>5.1 Zigzag machine is set up by ensuring the correct needle, thread, and bobbin.</p> <p>5.2 Stitch width and length are adjusted according to the fabric type.</p> <p>5.3 Sample fabric/cut piece is positioned under the presser foot.</p> <p>5.4 Machine is operated at a steady speed, maintaining consistent stitch formation.</p> <p>5.5 Tension is monitored and adjusted without puckering or thread breakage.</p> <p>5.6 During sewing, unusual conditions (if any) are detected and reported to authority.</p> <p>5.7 Machine is cleaned and maintained regularly as per the manufacturer's guidelines.</p>                            |
| 6. Operate bar tack machine  | <p>6.1 Bar tack machine is set up by ensuring the correct needle, thread, and bobbin.</p> <p>6.2 Stitch length and tension are adjusted according to the fabric type and the specific bar tack requirements.</p> <p>6.3 Sample fabric/cut piece is positioned accurately under the machine's presser foot.</p> <p>6.4 Machine is operated at a controlled speed ensuring secure bar tack stitch at the designated area.</p> <p>6.5 Machine is monitored to ensure the bar tack is correctly formed, with no skipped stitches or thread breakage.</p> <p>6.6 During sewing, unusual conditions (if any) are detected and reported to authority.</p> |

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|--|-----------------------------------------------------------------------------------------------|
|  | 6.7 Machine is cleaned and maintained regularly according to the manufacturer's instructions. |
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### Range of Variables

| Variable              | Range (Includes but not limited to):                                                                                                                |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Unusual conditions | 1.1 Excess oil/lubricant<br>1.2 Unusual needle breakage<br>1.3 Unusual thread breakage<br>1.4 Irregular stitch tension<br>1.5 Unusual machine noise |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Functions of different sewing machines including single needle lockstitch, double needle lockstitch, overlock, flat lock, zigzag, and bar tack machines.<br>1.2 Types and use of needle.<br>1.3 Explain the importance of adjusting thread, stitch length, stitch width, and tension according to fabric type and stitching requirements.<br>1.4 Understand position of fabric/cut piece.<br>1.5 Reasons of puckering or thread breakage.<br>1.6 Recognize and report unusual machine conditions.<br>1.7 Cleaning procedure of sewing machine. |
| 2. Underpinning Skills    | 2.1 Setting sewing machines including single needle lockstitch, double needle lockstitch, overlock, flat lock, zigzag, and bar tack machines.<br>2.2 Threading procedure.<br>2.3 Setting needle and stitches per inch<br>2.4 Controlling machine speed<br>2.5 Operating single needle lock stitch machine<br>2.6 Operating double needle lock stitch machine<br>2.7 Operating over lock machine<br>2.8 Operating flat lock machine<br>2.9 Operating zigzag machine<br>2.10 Operating bar tack machine                                              |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace                                                                                                                                                                                |
| 4. Resource Implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Tools, equipment and facilities appropriate to the                                                                                                                                                                                                                                                                                                                                                                                                         |

|  |                                                      |
|--|------------------------------------------------------|
|  | process or activities.                               |
|  | 4.3 Materials are relevant to the proposed activity. |
|  | 4.4 Specification and job card.                      |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <ol style="list-style-type: none"> <li>1.1 set and operated single needle lock stitch machine.</li> <li>1.2 set and operated double needle lock stitch machine.</li> <li>1.3 set and operated over lock machine.</li> <li>1.4 set and operated flat lock machine.</li> <li>1.5 set and operated zigzag machine.</li> <li>1.6 set and operated bar tack machine.</li> <li>1.7 adjusted stitch length and tension.</li> <li>1.8 detected unusual conditions.</li> <li>1.9 cleaned and maintained workplace.</li> </ol> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <ol style="list-style-type: none"> <li>2.1 Written test</li> <li>2.2 Practical Demonstration</li> <li>2.3 Oral Questioning</li> <li>2.4 Portfolio (Optional)</li> </ol>                                                                                                                                                                                                                                                                                                                                                             |
| 3. Context of Assessment          | <ol style="list-style-type: none"> <li>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</li> <li>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li> </ol>                                                                                                                                                                                                                                                        |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>PERFORM LINGERIE PRODUCT SEWING</b>                                                                                                                                                                                                  | <b>Nominal Duration:</b><br>62 Hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-04-O |
| <b>Unit Descriptor:</b><br>This unit covers the knowledge, skills and attitudes required to perform lingerie products sewing. It specifically includes the tasks of collecting bra and brief components, carrying out brief stitching and carrying out bra stitching. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency               | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Collect bra and brief components. | 1.1 <b><u>Customer/ buyer requirements</u></b> are identified.<br>1.2 <b><u>Components of bra and brief</u></b> are identified as per sample.<br>1.3 Bra and brief making process is identified.<br>1.4 Pattern for both the bra and brief is selected, ensuring design specifications.<br>1.5 Bra and brief components are arranged according to the pattern.                                                                                                                                                                                                                                                                                                                          |
| 2. Carry out brief stitching         | 2.1. Appropriate fabric and pattern for the brief are selected, ensuring the design and size requirements are met.<br>2.2. Fabric pieces for the front, back, and gusset are prepared according to the pattern.<br>2.3. Pieces are aligned and basted, ensuring accurate seam placement.<br>2.4. Stitching machine is set up, and the fabric is sewn using the appropriate stitch type.<br>2.5. Waistband and leg openings are finished with elastic or trim.<br>2.6. Brief is checked for any stitching errors.<br>2.7. Machine is cleaned and maintained regularly according to the manufacturer's instructions.                                                                      |
| 3. Carry out bra stitching           | 3.1 Appropriate fabric, pattern, and size for the bra are selected.<br>3.2 Components are selected according to the pattern.<br>3.3 Pieces are basted together to ensure correct alignment before stitching.<br>3.4 Stitching machine is set up with the correct thread and tension for smooth and durable seams.<br>3.5 Bra's cups, under bust, and shoulder straps are sewn together using the correct stitch type to ensure structural integrity and comfort.<br>3.6 Elastic is applied to the under bust, straps, and edges of the cups for a secure and flexible fit.<br>3.7 Bra is checked for any stitching errors.<br>3.8 Machine is cleaned and maintained regularly according |

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|  | to the manufacturer's instructions. |
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### Range of Variables

| Variable                       | Range (Includes but not limited to):                                                                                                                                                                                                                                                                   |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Customer/buyer requirements | 1.1 Design<br>1.2 Stitch per Inch (SPI)<br>1.3 Raw Edge (mm)<br>1.4 Needle Gauge<br>1.5 Thread type and color                                                                                                                                                                                          |
| 2. Components of bra and brief | 2.1 <b>Bra</b><br>2.1.1 Front strap<br>2.1.2 Back strap<br>2.1.3 Cups<br>2.1.4 Wings<br>2.1.5 Center gore/yoke<br>2.1.6 Hook & eye<br>2.1.7 Wires<br>2.1.8 Slider<br>2.1.9 Ring<br>2.2 <b>Brief</b><br>2.2.1 Waist band<br>2.2.2 Front part<br>2.2.3 Back part<br>2.2.4 Leg band<br>2.2.5 Gusset pouch |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Recognize customer/buyer's requirements.<br>1.2 Describe components of bra and brief.<br>1.3 Process flow chart of making bra and brief.<br>1.4 Understand pattern and specification's.<br>1.5 Explain stitch machine setup.<br>1.6 Stitching operation for bra and brief.<br>1.7 Assembling operation of bra and of brief<br>1.8 Cleaning procedure of sewing machine<br>1.9 Stitch quality. |
| 2. Underpinning Skills    | 2.1 Following OSH.<br>2.2 Interpreting design specifications of bra and brief.<br>2.3 Cleaning and checking sewing machine<br>2.4 Performing stitching operation of bra<br>2.5 Performing assembling operation of bra<br>2.6 Performing stitching operation of brief<br>2.7 Performing assembling operation of brief<br>2.8 Checking stitching quality                                            |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities                                                                                                                                                                                                                                                                                                     |

|                          |                                                                                                                                                                                                                                                                    |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace |
| 4. Resource Implications | The following resources must be provided:<br>4.1 Tools, equipment and facilities appropriate to the process or activities.<br>4.2 Materials are relevant to the proposed activity.<br>4.3 Design and specification.<br>4.4 Manuals.                                |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 identified customer /buyer requirements<br>1.2 identified bra and brief making process.<br>1.3 selected pattern for both the bra and brief.<br>1.4 carried out stitching of brief.<br>1.5 carried out stitching of bra.<br>1.6 cleaned and maintained machine and workplace. |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                                                                                             |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.                                                                        |

|                                                                                                                                                                                                                                                                                |                                     |                                          |
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| <b>Unit of Competency:</b><br><b>MAINTAIN OPTIMUM PRODUCTION.</b>                                                                                                                                                                                                              | <b>Nominal Duration:</b><br>20 Hrs. | <b>Unit Code:</b><br>SICIP-RMG-LSMO-05-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to maintain optimum production. It specifically includes the tasks of maintaining production target and efficiency, maintaining product quality and performing basic maintenance. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                       | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Maintain production target and efficiency | 1.1 Production target is clearly understood and communicated to the team.<br>1.2 Work processes are organized to optimize efficiency.<br>1.3 Resources are managed effectively to prevent delays and minimize wastage.<br>1.4 Any issues impacting efficiency are promptly addressed to minimize downtime.                                                                               |
| 2. Maintain products quality                 | 2.1. Product quality standards are defined.<br>2.2. Materials and components used in production are inspected.<br>2.3. <b><u>Defects</u></b> in products are identified to maintain overall product quality.<br>2.4. Final products are thoroughly checked for defects.                                                                                                                  |
| 3. Perform basic maintenance                 | 3.1 Basic maintenance schedules are followed.<br>3.2 <b><u>Preventive maintenance activities</u></b> are identified.<br>3.3 Safety guidelines are followed during maintenance procedures.<br>3.4 Necessary adjustments are performed to ensure smooth operation.<br>3.5 Minor repairs are carried out promptly to prevent equipment malfunction.<br>3.6 Tools and machinery are cleaned. |

### Range of Variables

| Variable   | Range (Includes but not limited to):                                                                                                                        |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Defects | 1.1 <b><u>Stitch Defects</u></b><br>1.1.1 Gathering<br>1.1.2 Broken stitch<br>1.1.3 Puckering<br>1.1.4 Open seam<br>1.1.5 Loose stitch<br>1.1.6 Skip stitch |

|                                      |                                                                                                                                                                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | 1.1.7 Floating stitch<br>1.1.8 Uneven stitch<br>1.1.9 Slipped stitch<br>1.1.10 Pleat stitch<br>1.1.11 Needle hole<br>1.2 <b>Other Defects</b><br>1.2.1 Oil spot<br>1.2.2 Size mixing<br>1.2.3 Stitch tension loose<br>1.2.4 Label missing<br>1.2.5 Wrinkles |
| 2. Preventive maintenance activities | 2.1. Feed dog check<br>2.2. Needle check<br>2.3. Needle plate check<br>2.4. Belt cover check<br>2.5. Pully cover check<br>2.6. Needle guard check<br>2.7. Fabric guide check<br>2.8. Eye glass check                                                        |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Understand production target.<br>1.2 Importance of productivity<br>1.3 Resources management.<br>1.4 Impact of efficiency.<br>1.5 Products quality.<br>1.6 Explain defects in products.<br>1.7 Describe preventive maintenance activities.<br>1.8 Safety guidelines of maintenance.                                                                              |
| 2. Underpinning Skills    | 2.1. Ensuring production target.<br>2.2. Checking products quality<br>2.3. Inspecting production materials and components.<br>2.4. Maintaining products quality.<br>2.5. Checking products defects.<br>2.6. Performing preventive maintenance activities.<br>2.7. Cleaning tools and machinery.                                                                     |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace |
| 4. Resource Implications  | The following resources must be provided:                                                                                                                                                                                                                                                                                                                           |



|  |                                                                                                                                                                                                                    |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 4.1 Work place (Simulated)<br>4.2 Tools, equipment and facilities appropriate to the process or activities.<br>4.3 Materials are relevant to the proposed activity.<br>4.4 Buyer's specifications.<br>4.5 Manuals. |
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### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 Maintained production target and efficiency.<br>1.2 Identified defects in products.<br>1.3 Maintained products quality.<br>1.4 Performed basic maintenance.                                           |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                      |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

### Workshop/Lab Facility Standard

|                            |                                     |
|----------------------------|-------------------------------------|
| <b>Course Name:</b>        | Sewing Machine Operation (Lingerie) |
| <b>Number of Trainees:</b> | 30                                  |

**Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):**

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)  
OR
- Classroom cum workshop – 1000 sft (93 sqm)

**Major Training Equipment and Training Facilities:**

| S.N. | Major Equipment and Training facilities | Required facilities |
|------|-----------------------------------------|---------------------|
| 1.   | Computers/ Laptop                       | 01                  |
| 2.   | Multimedia projector screen             | 01                  |
| 3.   | Single needle lock stitch machine       | 30                  |
| 4.   | Double needle lockstitch machine        | 02                  |
| 5.   | Over Lock Machine                       | 04                  |
| 6.   | Flat Lock Machine                       | 04                  |
| 7.   | Single step zigzag machine              | 02                  |
| 8.   | 3 step zigzag machine                   | 02                  |
| 9.   | Bar Tack Machine                        | 01                  |
| 10.  | Measuring Tape                          | 25                  |
| 11.  | Stop Watch                              | 02                  |
| 12.  | Calculator                              | 02                  |
| 13.  | Fabric Cutting Scissor (6-10 inches)    | 05                  |
| 14.  | Fabric Cutting Table (Min. 5 Feet)      | 01                  |
| 15.  | Steam Iron Set                          | 02                  |
| 16.  | Steam Iron Table                        | 02                  |
| 17.  | Fabric Cutting Table (Min. 5 Feet)      | 1                   |
| 18.  | Finishing Table (Min. 5 Feet)           | 1                   |
| 19.  | Digital Video Camera                    | 01                  |
| 20.  | Printer                                 | 01                  |
| 21.  | White board                             | 01                  |

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Sewing Machine Operation (Lingerie) the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 30 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 30 trainees.